

Experiment No.	8
Experiment Title	Clustering Human Activity Recognition Data using K-Means, DBSCAN, and Hierarchical Clustering
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1 Objective

The objective of this experiment is to implement and compare different clustering algorithms on the Human Activity Recognition dataset.

The following three clustering algorithms are used:

1. K-Means Clustering
2. DBSCAN
3. Hierarchical Agglomerative Clustering

The clusters formed by these algorithms are visualized and compared using different clustering evaluation metrics.

2 Problem Statement

The task is to group the Human Activity Recognition data into clusters without using the activity labels during clustering.

K-Means is used with different values of k and the Elbow method and Silhouette Score are used to select the number of clusters. DBSCAN is tested with different values of ϵ and `min_samples`. Hierarchical Agglomerative Clustering is then applied using Ward's linkage.

The final clusters are compared using internal metrics such as Silhouette Score, Davies-Bouldin Index, and Calinski-Harabasz Score. ARI and NMI are also used to compare the clusters with the actual activity labels.

3 Dataset Description

The dataset used in this experiment is the Human Activity Recognition Using Smartphones dataset from the UCI Machine Learning Repository.

Dataset Name	Human Activity Recognition Using Smartphones
Dataset Source	UCI Machine Learning Repository
Number of Samples	10,299
Number of Features	561
Number of Activities	6

Missing Values	0
Duplicate Rows	0

The six activities in the dataset are:

1. WALKING
2. WALKING_UPSTAIRS
3. WALKING_DOWNSTAIRS
4. SITTING
5. STANDING
6. LAYING

The dataset was collected from 30 volunteers using smartphone accelerometer and gyroscope sensors. The sensor signals were processed into feature values for different human activities.

The distribution of the activities in the dataset is shown below.

Table 2: Activity Distribution

Activity	Number of Samples
LAYING	1944
STANDING	1906
SITTING	1777
WALKING	1722
WALKING_UPSTAIRS	1544
WALKING_DOWNSTAIRS	1406

4 Exploratory Data Analysis

Exploratory Data Analysis (EDA) was performed before applying the clustering algorithms.

First, the training and testing feature data were combined, giving a total of 10,299 samples and 561 features.

The dataset was also checked for missing values and duplicate rows. No missing values or duplicate rows were found.

The activity labels were mapped to their corresponding activity names and their distribution was checked.

Since the dataset has 561 features, it is difficult to directly visualize the data. PCA was therefore used later to reduce the data to two dimensions for visualization.

5 Data Preprocessing

The following preprocessing steps were performed:

1. The training and testing feature files were loaded.
2. The training and testing label files were loaded.
3. The training and testing data were combined.
4. Feature names were assigned using the provided feature file.
5. The activity numbers were mapped to activity names.
6. Missing values and duplicate rows were checked.
7. The features were standardized using `StandardScaler`.

After preprocessing, the shape of the scaled data was:

(10299, 561)

Standardization was used because the clustering algorithms are based on distances and features with different scales can affect the clustering result.

6 Principal Component Analysis

PCA was used to visualize the high-dimensional data in two dimensions.

The original dataset contains 561 features, so directly plotting the data is not possible. PCA was used to transform the data into two principal components.

The first principal component explained approximately 50.74% of the variance, while the second component explained approximately 6.24%.

Therefore, the first two components together explained approximately:

56.98%

of the total variance.

PCA was mainly used for visualization in this experiment. The clustering algorithms were applied to the standardized feature data.

7 Clustering Algorithms

7.1 K-Means Clustering

K-Means divides the data into a fixed number of clusters. Each point is assigned to the nearest centroid and the centroids are updated until the clusters become stable.

The objective of K-Means is to minimize the within-cluster sum of squares:

$$WCSS = \sum_{i=1}^k \sum_{x \in C_i} \|x - \mu_i\|^2$$

Different values of k were tested from 2 to 7.

The Elbow method was used to observe the change in WCSS and the Silhouette Score was used to select the final value of k .

7.2 DBSCAN

DBSCAN is a density-based clustering algorithm. Unlike K-Means, it does not require the number of clusters to be given beforehand.

The main parameters used were:

- ϵ – radius of the neighbourhood.
- `min_samples` – minimum number of points required to form a dense region.

DBSCAN can also mark points as noise when they do not belong to any dense region. Different combinations of ϵ and `min_samples` were tested.

7.3 Hierarchical Agglomerative Clustering

Hierarchical clustering starts with each observation as a separate cluster and then repeatedly merges the closest clusters.

Ward linkage was used in this experiment. The number of clusters was selected as 2 based on the K-Means result.

A dendrogram was also plotted using a subset of the data to make the hierarchical structure easier to visualize.

8 Experimental Setup

Dataset Size	10,299 samples, 561 features
Scaling	StandardScaler
PCA	2 components for visualization
K-Means Range	$k = 2$ to 7
DBSCAN ϵ	1 to 8
DBSCAN min samples	5, 10, 15
Hierarchical Method	Agglomerative, Ward linkage

The actual activity labels were not used while forming the clusters. They were only used later to compare the clustering results with the known activities.

9 Evaluation Metrics

The clustering results were evaluated using the following metrics.

9.1 Silhouette Score

Silhouette Score measures how close a point is to its own cluster compared with the other clusters. A higher value generally means that the clusters are better separated.

9.2 Davies-Bouldin Index

Davies-Bouldin Index measures the similarity between clusters. A lower value indicates better separation between clusters.

9.3 Calinski-Harabasz Score

Calinski-Harabasz Score compares the separation between clusters with the spread within each cluster. A higher value indicates better clustering.

9.4 Adjusted Rand Index

ARI compares the generated clusters with the actual activity labels while taking random agreement into account. A higher value means better agreement with the known labels.

9.5 Normalized Mutual Information

NMI measures the similarity between the generated clusters and the actual activity labels. Higher values indicate better agreement.

10 K-Means Results

K-Means was run for different values of k from 2 to 7.

Table 3: K-Means Results		
k	WCSS	Silhouette Score
2	3272857	0.3937
3	2921078	0.3155
4	2781602	0.1500
5	2654790	0.1286
6	2577180	0.1099
7	2519774	0.0816

The highest Silhouette Score was obtained for $k = 2$:

$$k = 2$$

with a Silhouette Score of:

$$0.3937$$

The two clusters contained 5620 and 4679 samples respectively.

The Silhouette Score decreased as the value of k increased. This shows that using more clusters did not improve the separation of the data for this dataset.

11 DBSCAN Results

Different combinations of ϵ and `min_samples` were tested for DBSCAN.

The selected configuration was:

$$\epsilon = 8, \quad \text{min_samples} = 5$$

This configuration produced 5 clusters.

However, most of the observations were classified as noise. The result contained:

$$10272 \text{ noise points}$$

out of 10,299 observations.

This means that DBSCAN was not able to group most of the observations into dense clusters with the selected parameters.

The Silhouette Score for the selected DBSCAN result was 0.3650.

12 Hierarchical Clustering Results

Hierarchical Agglomerative Clustering was applied using Ward linkage with 2 clusters.

The resulting cluster sizes were:

Table 4: Hierarchical Cluster Sizes

Cluster	Number of Samples
Cluster 0	4672
Cluster 1	5627

The Silhouette Score obtained was 0.3934, which was very close to the K-Means result. A dendrogram was also generated using a subset of 500 samples.

13 Clustering Results Comparison

The final results of the three clustering algorithms are compared below.

Table 5: Comparison of Clustering Algorithms

Algorithm	Silhouette	DB	CH	ARI	NMI
K-Means	0.3937	1.0707	7880.8139	0.3296	0.5455
DBSCAN	0.3650	0.9717	14.5078	0.1752	0.4611
Hierarchical	0.3934	1.0717	7859.7554	0.3325	0.5568

Here, DB represents Davies-Bouldin Index and CH represents Calinski-Harabasz Score. K-Means gave the highest Silhouette Score of 0.3937. Hierarchical clustering was very close with a score of 0.3934.

Hierarchical clustering had slightly better ARI and NMI values than K-Means, which means its clusters had slightly better agreement with the actual activity labels.

DBSCAN had the lowest ARI and NMI values and also classified most of the samples as noise.

14 Result Visualization

The following plots were generated in the notebook:

1. Activity distribution plot.
2. Feature distribution plots.
3. PCA 2D visualization.
4. K-Means Elbow curve.
5. K-Means Silhouette Score curve.
6. K-Means cluster visualization.
7. DBSCAN cluster visualization.
8. Hierarchical cluster visualization.
9. Hierarchical clustering dendrogram.
10. Comparison of clustering metrics.

14.1 Activity Distribution

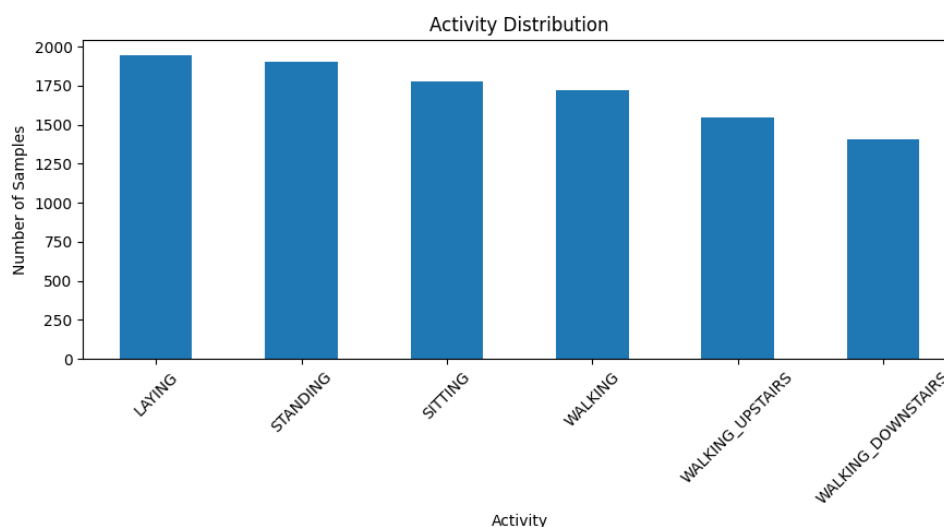


Figure 1: Distribution of the six human activities.

The six activities have different numbers of observations, with LAYING having the highest number of samples and WALKING_DOWNSTAIRS having the lowest.

14.2 PCA Visualization

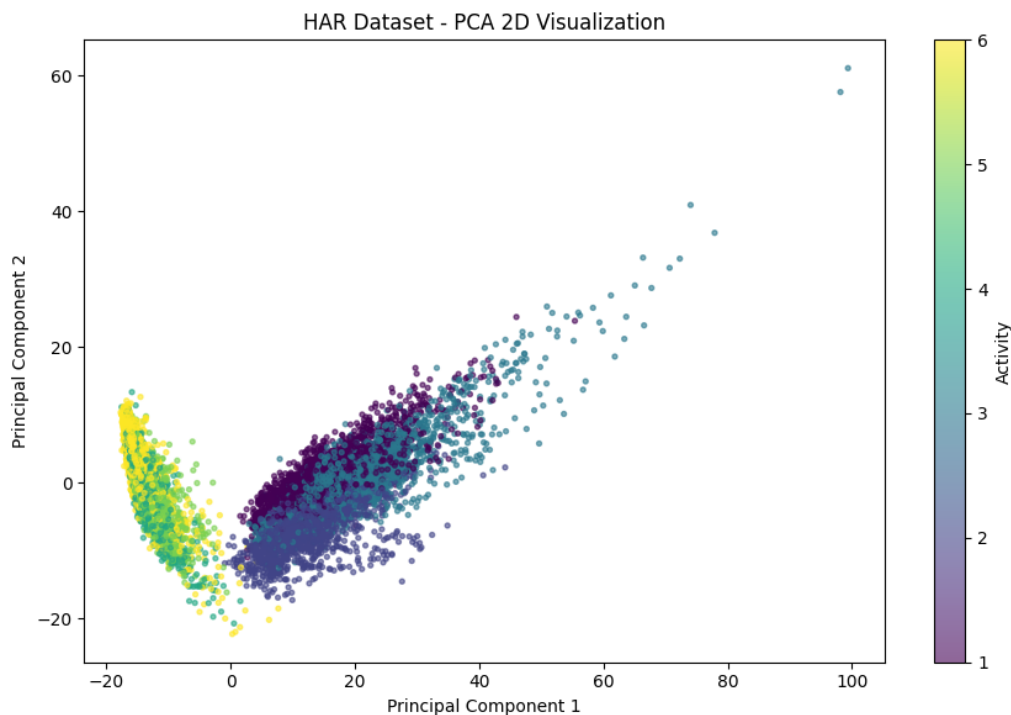


Figure 2: PCA visualization of the Human Activity Recognition dataset.

PCA was useful for getting a visual idea of how the activities are distributed in a two-dimensional space.

14.3 K-Means Elbow Curve

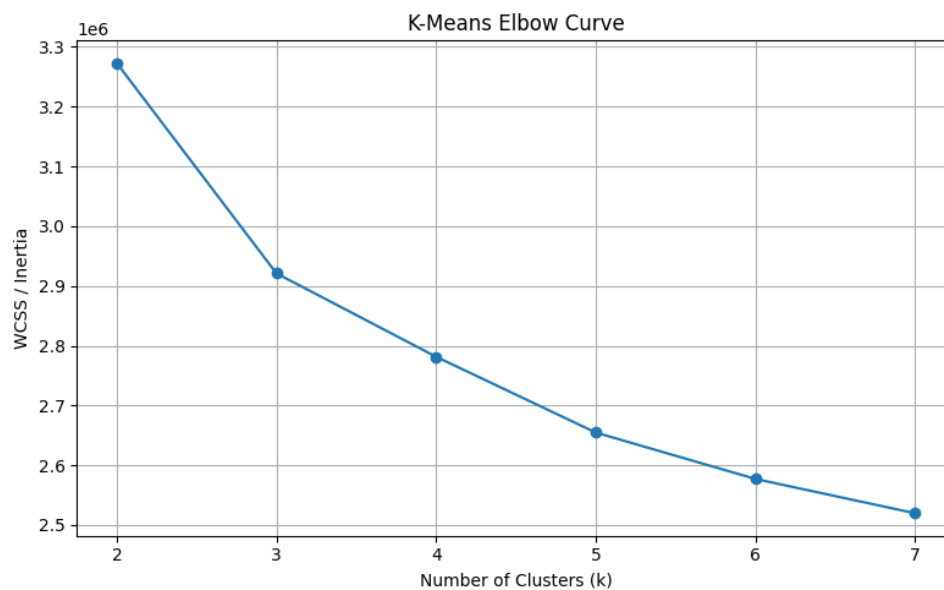


Figure 3: Elbow curve for K-Means.

The WCSS decreased as k increased. The Silhouette Score was also considered when selecting the final number of clusters.

14.4 K-Means Silhouette Curve

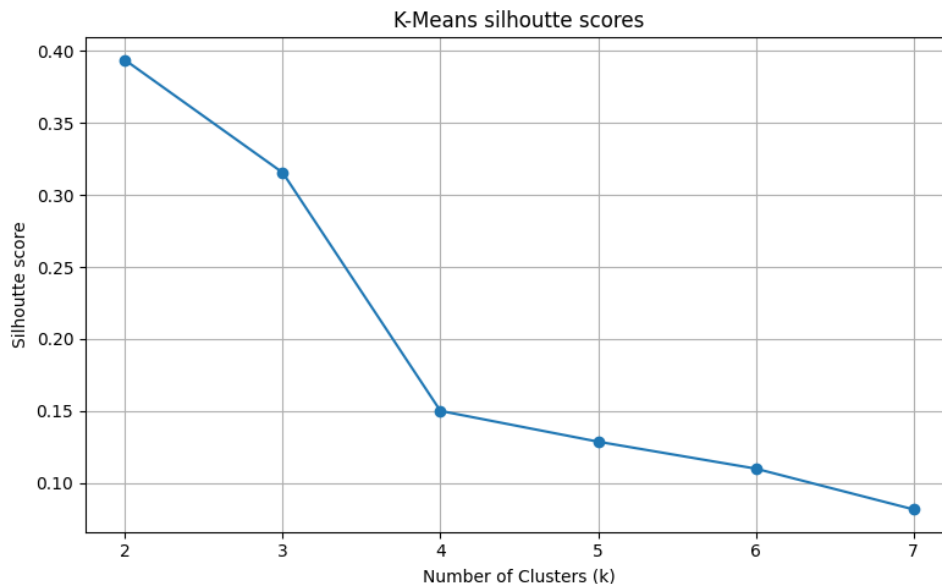


Figure 4: Silhouette Score for different values of k .

The highest Silhouette Score was obtained at $k = 2$.

14.5 K-Means Clusters

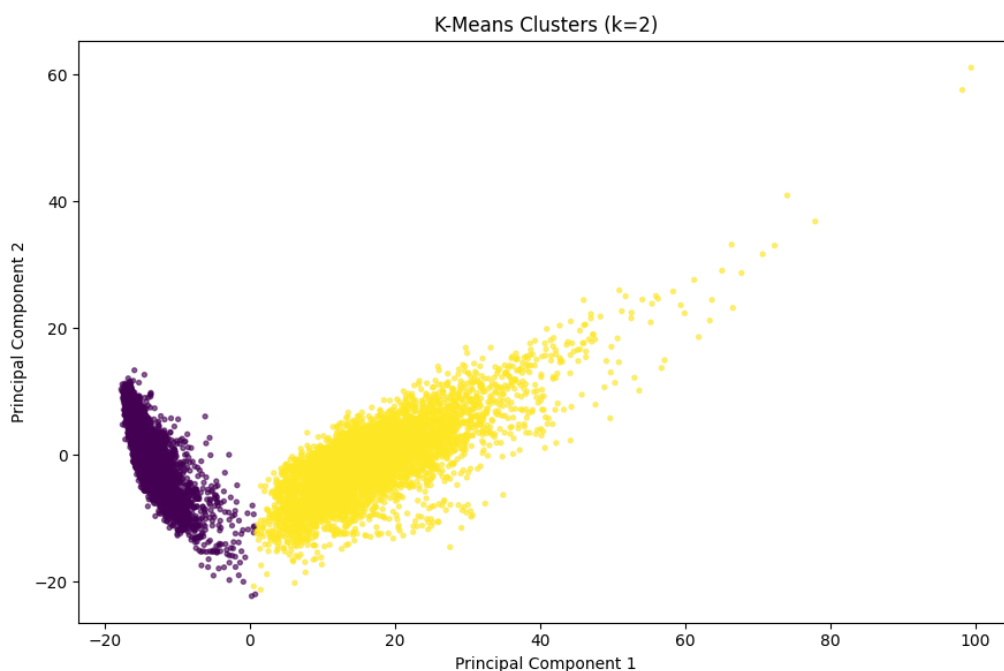


Figure 5: K-Means clusters shown using PCA components.

14.6 DBSCAN Clusters

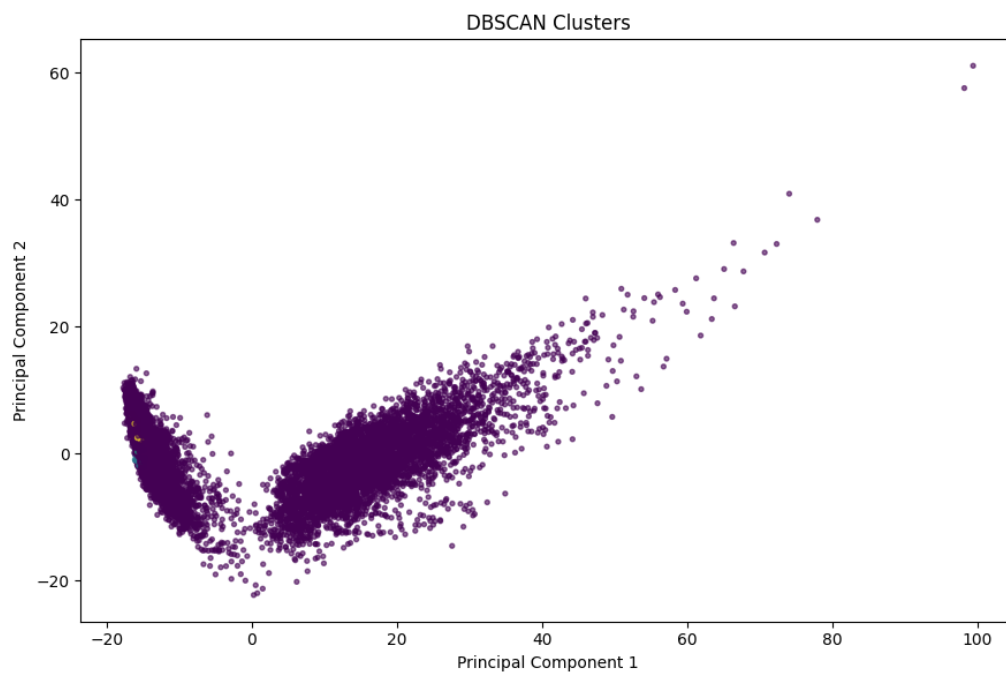


Figure 6: DBSCAN clusters shown using PCA components.

The DBSCAN plot shows that a large number of observations were treated as noise.

14.7 Hierarchical Clusters

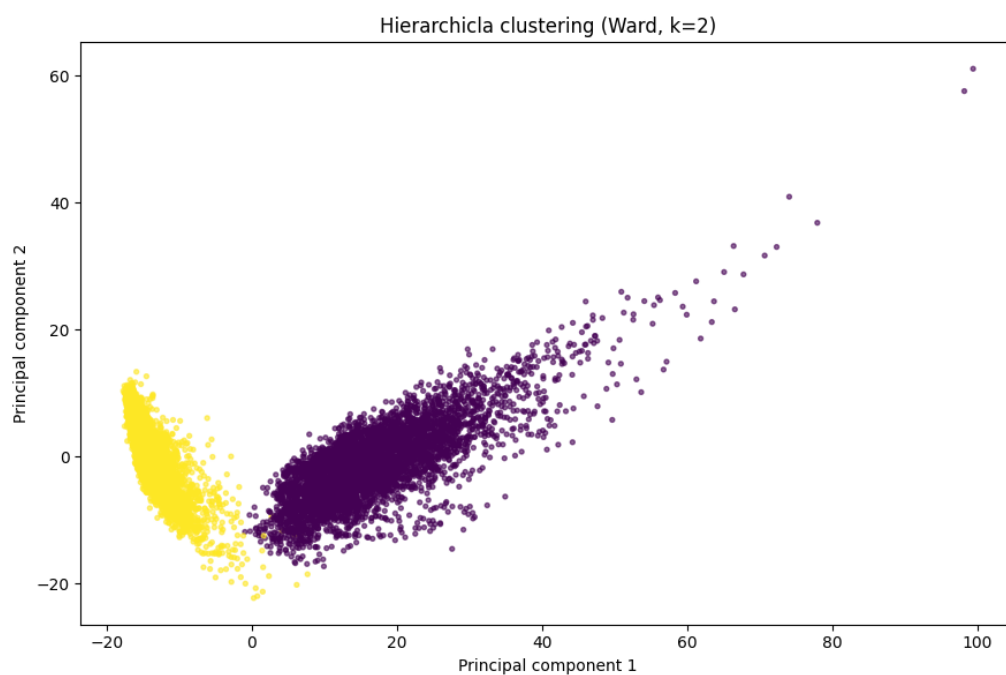


Figure 7: Hierarchical clustering using Ward linkage.

14.8 Dendrogram

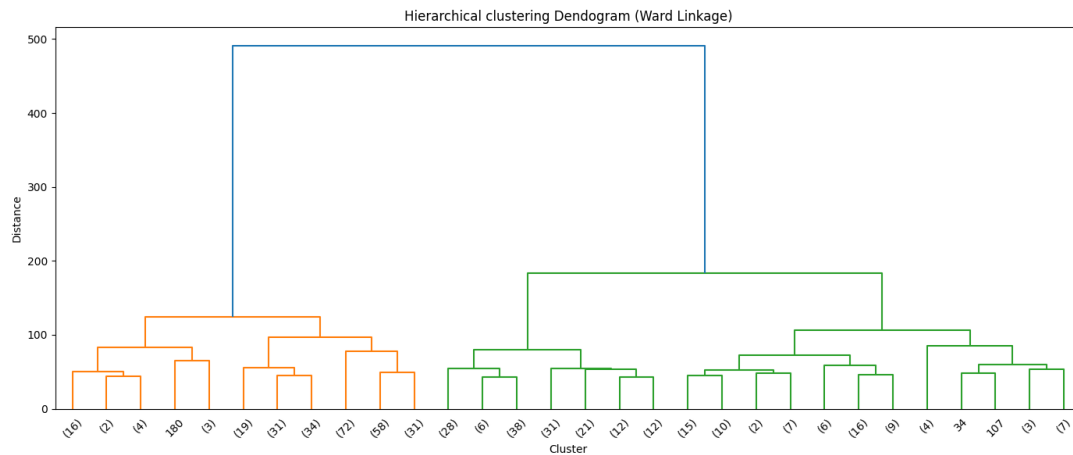


Figure 8: Dendrogram for hierarchical clustering.

The dendrogram was generated using 500 samples so that the result would be easier to view.

15 Discussion

The results show that K-Means and Hierarchical clustering performed similarly on this dataset.

K-Means gave the highest Silhouette Score of 0.3937, while Hierarchical clustering gave 0.3934. Their Calinski-Harabasz Scores were also very close.

Hierarchical clustering had a slightly higher ARI and NMI than K-Means. However, the difference was small.

DBSCAN produced a lower Silhouette Score and had much lower ARI and NMI values. The main problem with the DBSCAN result was that 10,272 samples were classified as noise.

K-Means was also sensitive to the choice of k . The Silhouette Score was highest at $k = 2$ and decreased as more clusters were added.

Overall, K-Means and Hierarchical clustering gave more useful results for this dataset compared to the DBSCAN configuration tested.

16 Conclusion

This experiment compared K-Means, DBSCAN, and Hierarchical clustering on the Human Activity Recognition dataset containing 10,299 samples and 561 features.

The data was standardized before clustering and PCA was used to visualize the data in two dimensions.

K-Means achieved the highest Silhouette Score of 0.3937 with $k = 2$. Hierarchical clustering gave a very similar Silhouette Score of 0.3934 and slightly better ARI and NMI values.

DBSCAN produced 5 clusters, but 10,272 samples were classified as noise. Therefore, it was not as effective for this dataset with the tested parameters.

Overall, K-Means and Hierarchical clustering gave similar and more useful results, while DBSCAN was affected by the density of the high-dimensional data.

17 References

1. UCI Machine Learning Repository, “Human Activity Recognition Using Smartphones Dataset.”
2. UCI Machine Learning Repository, <https://archive.ics.uci.edu/dataset/240/human%2Bactivity%2Brecognition%2Busing%2Bsmartphones>
3. Scikit-learn Documentation, <https://scikit-learn.org/>
4. MacQueen, J., “Some Methods for Classification and Analysis of Multivariate Observations,” Proceedings of the Fifth Berkeley Symposium on Mathematical Statistics and Probability, 1967.
5. Ester, M., Kriegel, H.-P., Sander, J., and Xu, X., “A Density-Based Algorithm for Discovering Clusters in Large Spatial Databases with Noise,” Proceedings of the Second International Conference on Knowledge Discovery and Data Mining, 1996.
6. Ward, J. H., “Hierarchical Grouping to Optimize an Objective Function,” Journal of the American Statistical Association, 1963.